<u>Do Stablecoins Predict Cryptos? A Copula-based approach</u>

Overview

We used 5 years of daily stablecoin and crypto data (2020-2024) to assess if stablecoins carry predictive power for major crypto market dynamics.

We used the following data and metrics:

Coin Type	Coins Used	Metrics Used
Stablecoin	- USDT - USDC - DAI	- Delta Log RV - Log Volume % Change
Major Crypto	- BTC - ETH - XRP - BNB	- Delta Log RV - Log Returns

The first thing to consider is how we determine which stablecoin metrics have predictive power for which crypto metrics

The naive approach, that many papers seem to use, is to test every stablecoin/crypto metric pair individually. For example, with 3 stablecoins, 2 stablecoin metrics, 4 cryptos, and 2 crypto metrics, we would have to run 48 individual tests. This creates two issues:

- Stablecoin metrics are highly correlated with each other, and crypto metrics are highly correlated with each other. For example, seeing if USDT volatility predicts BTC returns is an extremely similar test to seeing if USDC volatility predicts BTC returns. Most of the 48 tests would essentially be repeats and give us no new information
- With 48 tests at significant level 0.05, we are guaranteed to fine false positives (48*0.05 = 2.4, so expect at least 2 false positives)

To overcome this, we use **Principal Component Analysis (PCA).** PCA is a dimensionality reduction technique that can convert multiple time series into a single series that captures the **maximum shared variance** across all time series (PC1). Using PCA, we convert all our time series into 4 time series:

- Stablecoin Volume
- Stablecoin Volatility
- Crypto Volatility
- Crypto Returns

This means we now must only perform 4 tests:

- Stablecoin Volume -> Crypto Volatility
- Stablecoin Volatility -> Crypto Volatility
- Stablecoin Volume -> Crypto Returns
- Stablecoin Volatility -> Crypto Returns

Each time series of a single coin is made of two parts:

- The systemic signal that affects the market as a whole (Captured by PCA)
- The idiosyncratic noise for that specific coin (could be a specific event for that coin)

PCA helps us strip the noise and leave us with the true, systemic signals.

Lastly, PCA helps to avoid **multicollinearity.** For example, in the case of a ML model, if we used each individual coin metric as a feature, the model would not know which one was important, causing its feature importance and coefficients to become unstable. PCA also offers interpretability by allowing us to examine the **factor loadings** to see which coins are most influential

Methodology

First, we test for stationarity in each of our time series. We do this before PCA as otherwise the generated PC1 series would tell us very little.

We conduct 3 main tests:

- Copula Granger Causality (In Sample)
- Copula – GARCH forecasting (OOS, LogLikelihood and VaR)
- Copula – ML forecasting (OOS, XGBoost, MSE and Sharpe)

The theme across each of these tests is the **copula**. A copula allows us to separate the dependence structure of a set of variables from their individual behaviours. In other words, it allows us to see how variables move together without considering their individual behaviours. In practice, this is creating a joint distribution between variables. A copula can capture non-linear relationships.

Using a copula moves the focus of the study from looking at average relationships to ‘shock-based’ relationships. Again, it does this by removing the ‘noise’ of coin’s individual behaviours and capturing their hidden dependencies on one another. This allows us to look at extreme/unexpected events. This is a far more realistic model as these unexpected events are often what drive risk analysis and potential trading opportunities.

For the OOS forecasting, we use the first 4 years of our dataset (2020-2023) to train the model and the final year (2024) to test it.

We now breakdown the purpose of each test and why we use a copula:

Test	**In/Out of Sample?**	**Purpose**	**Result (If significant)**	**Why Copula?**
Granger Causality	In sample	To first establish if a predictive link exists at all	Can show lagged stablecoin data has a relationship with current crypto data	Standard Granger Causality is linear; copula introduces non-linearity Copula also makes it ‘shock-based’ – testing if unexpected movements in stablecoins predicts unexpected movements in cryptos
GARCH	OOS	Parametric Test Forecasts a full distribution	Statistical Proof (LogLik AG test) and Economic Proof (VaR backtest) that Stablecoin carries predictive power	Standard method of incorporating Stablecoin data would be GARCH-X style model where you add it as an exogenous variable. Assumes a linear relationship and ignores tail-dependence Using a copula again allows for non-linear relationships and a ‘shock-based’ prediction model: “given yesterday’s stablecoin shock, what is the new adjust risk model for crypto today?”
ML (XGBoost)	OOS	Non-parametric test Forecasts a single value/point	Can show Statistical Proof (MSE) and Economic Proof (Sharpe backtest) that Stablecoin carries predictive power	Filtering our features through a GARCH model then fitting a copula to the resulting shocks gives us very ‘clean’ features to add The copula strips away the ‘noise’ between the features and captures their true relationship. (Still a WIP)

In the following table, we explain the specific methodologies of each test:

<table>
<tr><th>Test</th><th>Approach</th></tr>
<tr><td>Granger Causality</td><td>Follow the methodology outlined in the paper

“A copula approach to assessing Granger causality”

- Create systemic series using PCA
- Select optimal lag for each series (informed by ADF lags)
- Use Kernel Density Estimation to obtain the shocks of the two variables Stablecoin Predictor, Crypto Target)
- Use Bernstein Estimation to estimate the dependence structure between the two variables
- Calculates a granger causality statistic for the copula
- Use bootstrapping to create 1000 synthetic datasets, calculating the granger causality statistic for each
- Construct a null distribution of these 1000 statistics.
- If our statistic for the true dataset is stronger than 95% of these values, we have a significant result!</td></tr>
<tr><td>GARCH</td><td>Follow the methodology outlined in the paper

“The Copula-GARCH model of conditional dependencies: An international stock market application
”

This paper only covers the construction of the copula GARCH model, the forecasting is of our own design

- Create systemic series using PCA
- Fit a AR – GARCH model to each PCA series. The AR part models the autocorrelation in the PCA’s mean then the GARCH part models the shocks from the AR model. This ensures the shocks are ‘clean’ and uncorrelated
- Train a copula on the shocks of the stablecoin and crypto data
- Generate a 1-day-ahead distribution forecast for day t
- Move to day t+1, storing the true value for day t and retrain
- Compare our copula model with the standard GARCH model for total log likelihood over the 1 year test set. Use an Amisano-Giacomini (AG) test to see if the difference between models is statistically significant
- Compare our copula model with the standard GARCH model for VaR breach rates at the 1% and 5% levels. See which model was closer to the true breach rate.</td></tr>
</table>

ML (XGBoost)	- Create systemic series using PCA - Fit a AR – GARCH model to each PCA series to be left with the shocks (As in GARCH above) - Feed these shocks to the ML model (XGBoost). The model then learns the dependence structure (copula) between these shocks. - Generate a 1-day-ahead point forecast for day t - Move to day t+1, storing the true value for day t and retrain - Compare our copula model with the standard XGBoost model by comparing MSE values - Compare our copula model with the standard XGBoost model by comparing Sharpe Ratios for a simple portfolio/strategy

Results

ADF:

	Coin	Variable	ADF Statistic	p-value	Number of Lags	Result
0	DAI	LogVolChange	-11.354390	9.815838e-21	25	Stationary
1	USDC	LogVolChange	-10.531926	9.104852e-19	25	Stationary
2	USDT	LogVolChange	-10.678880	3.987702e-19	25	Stationary
3	DAI	Delta_LogRV	-24.617229	0.000000e+00	1	Stationary
4	USDC	Delta_LogRV	-24.693063	0.000000e+00	1	Stationary
5	USDT	Delta_LogRV	-20.184756	0.000000e+00	2	Stationary
6	BNB	Log Returns	-10.430515	1.615297e-18	13	Stationary
7	BTC	Log Returns	-13.946580	4.785238e-26	8	Stationary
8	ETH	Log Returns	-12.883503	4.605595e-24	9	Stationary
9	XRP	Log Returns	-44.081903	0.000000e+00	0	Stationary
10	BNB	Delta_LogRV	-18.498891	2.121882e-30	3	Stationary
11	BTC	Delta_LogRV	-18.279061	2.310522e-30	4	Stationary
12	ETH	Delta_LogRV	-18.261831	2.330169e-30	4	Stationary
13	XRP	Delta_LogRV	-26.127422	0.000000e+00	1	Stationary

Copula Granger Causality:

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Source_Factor,Target_Factor,lag,Copula GC,p-value
Stable_Volume,Crypto_Volatility,30,4.50725392533929,0.027
Stable_Volatility,Crypto_Volatility,30,4.506256482530616,0.027
Stable_Volume,Crypto_Returns,10,1.4210992429671316,0.699
Stable_Volatility,Crypto_Returns,10,1.4000803619736646,0.718
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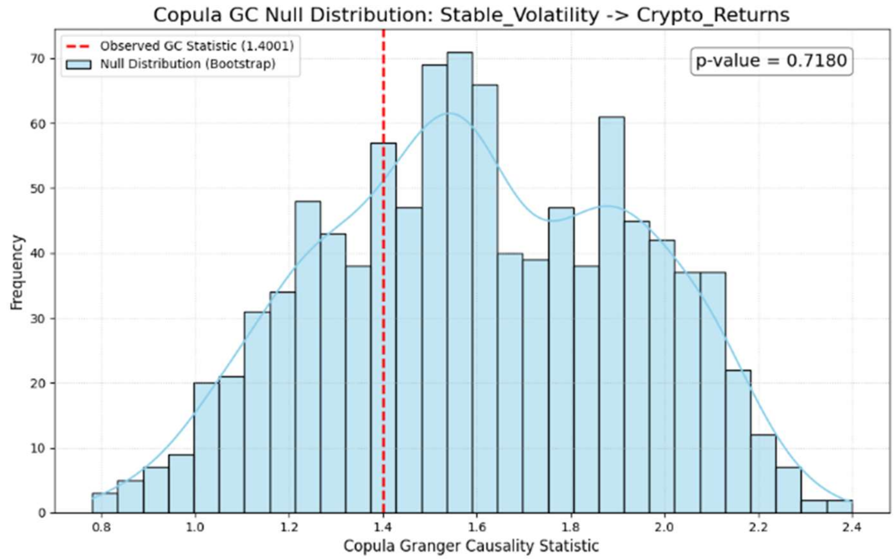
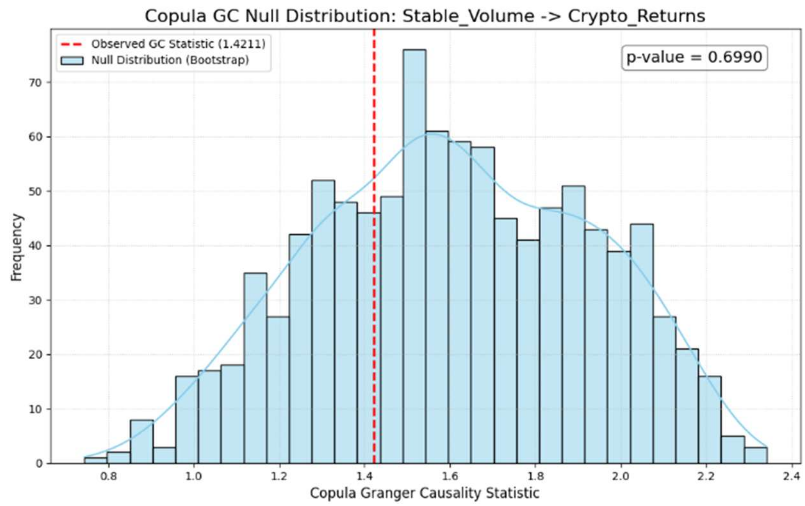
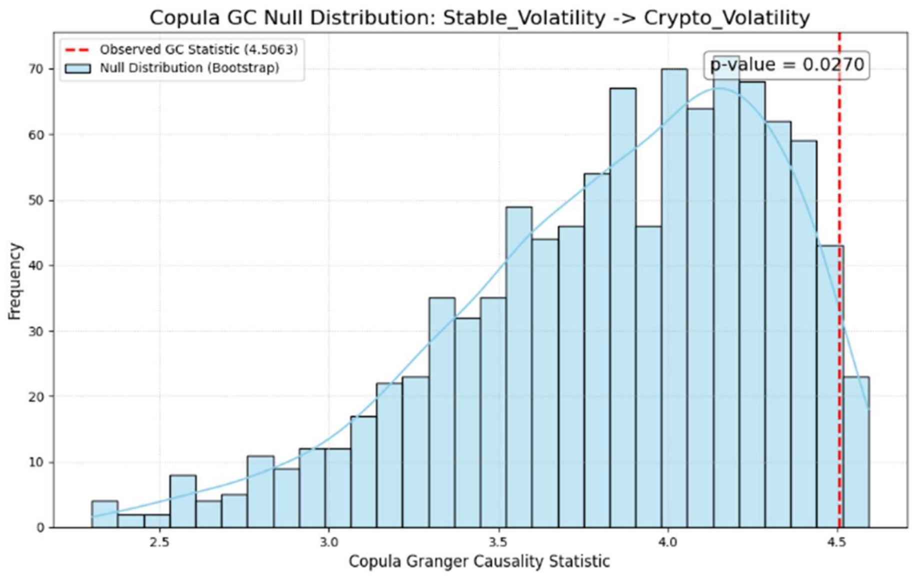
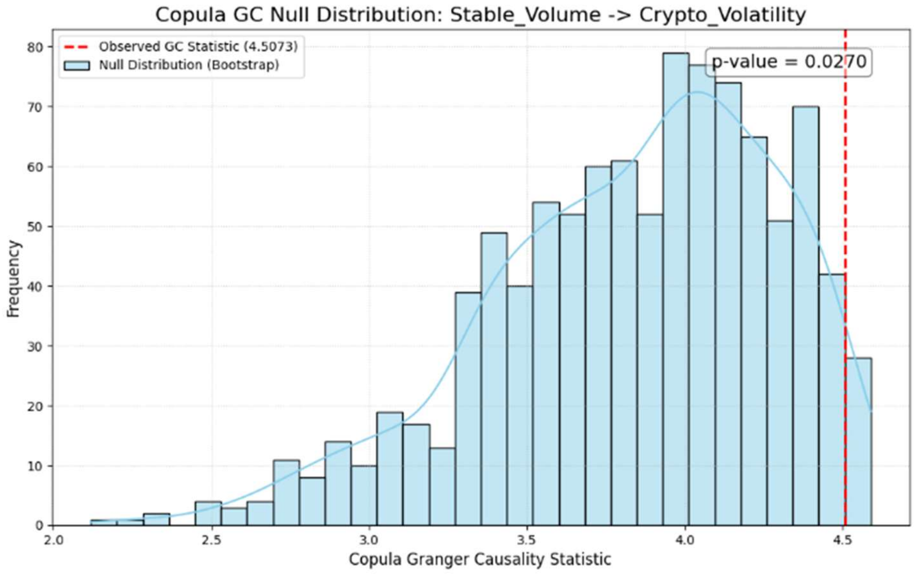

Copula GC Null Distribution: Stable_Volatility -> Crypto_Returns
Observed GC Statistic (1.4001)
Null Distribution (Bootstrap)
p-value = 0.7180
Frequency
Copula Granger Causality Statistic


Copula GC Null Distribution: Stable_Volume -> Crypto_Returns
Observed GC Statistic (1.4211)
Null Distribution (Bootstrap)
p-value = 0.6990
Frequency
Copula Granger Causality Statistic


Copula GC Null Distribution: Stable_Volatility -> Crypto_Volatility
Observed GC Statistic (4.5063)
Null Distribution (Bootstrap)
p-value = 0.0270
Frequency
Copula Granger Causality Statistic



GARCH (Log Likelihood):

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Final LAGGED OUT-OF-SAMPLE Amisano-Giacomini Test Results
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       Source_Factor     Target_Factor  OOS_Days  Mean_OOS_LL_Bench  Mean_OOS_LL_Copula_Part  AG_Statistic    AG_p_value
0       Stable_Volume    Crypto_Returns       366          -1.623760                -0.000737     -0.208939  8.346124e-01
1       Stable_Volume Crypto_Volatility       366          -2.154495                -0.008767     -1.737822  8.308563e-02
2   Stable_Volatility    Crypto_Returns       366          -1.623948                 0.085741      5.158317  4.096942e-07
3   Stable_Volatility Crypto_Volatility       363          -3.962236                 0.600210     11.923782  7.127936e-28
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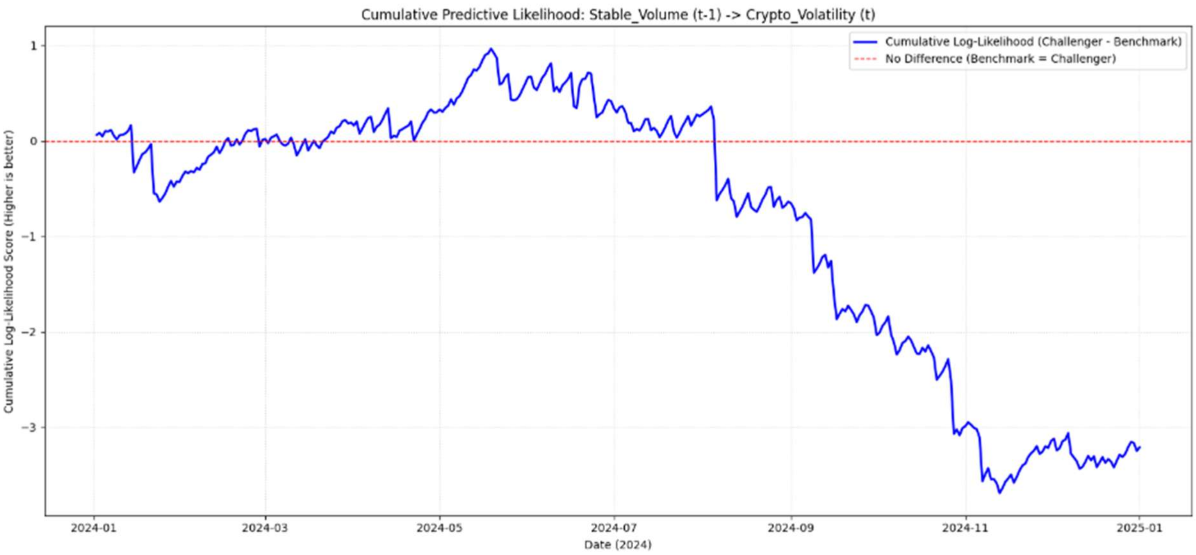
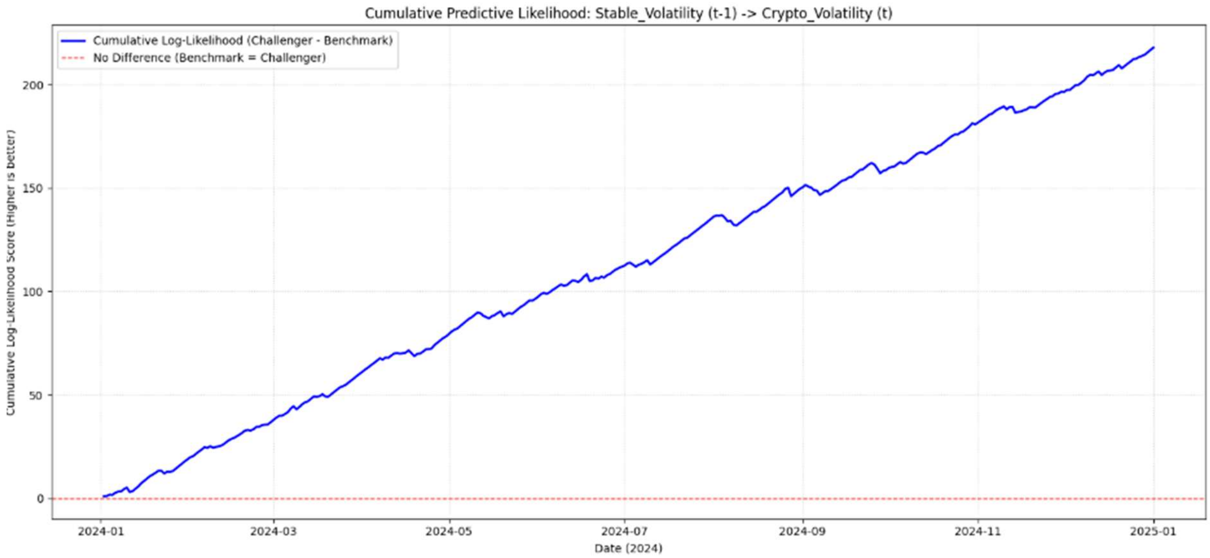
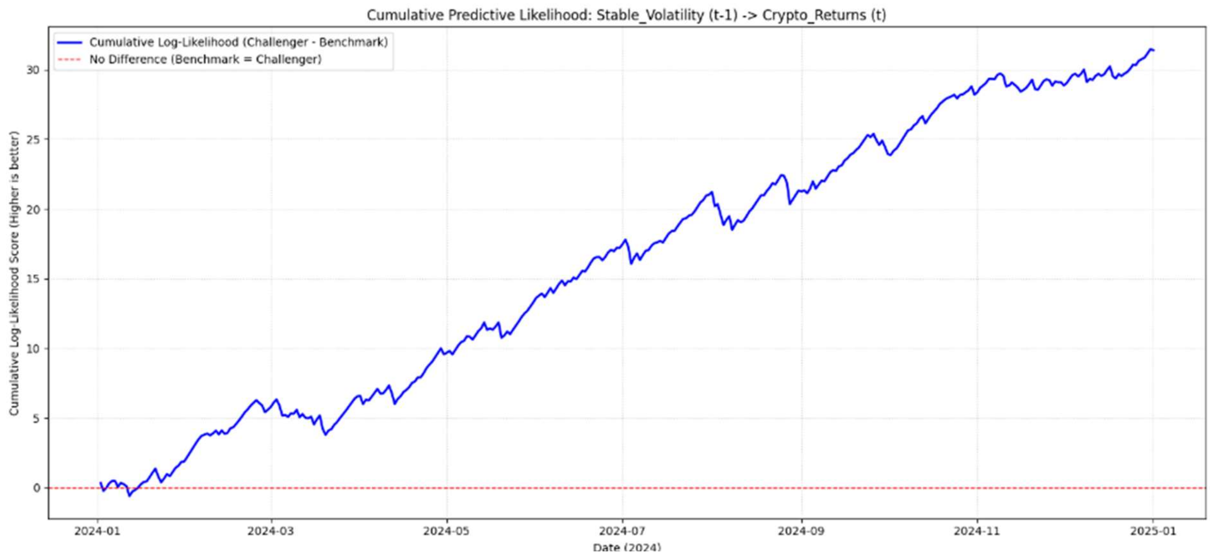



Cumulative Predictive Likelihood: Stable_Volume (t-1) -> Crypto_Returns (t)
Cumulative Log-Likelihood (Challenger - Benchmark)
No Difference (Benchmark = Challenger)
Cumulative Log-Likelihood Score (Higher is better)
Date (2024)
1.5
1.0
0.5
0.0
−0.5
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01


Cumulative Predictive Likelihood: Stable_Volatility (t-1) -> Crypto_Volatility (t)
Cumulative Log-Likelihood (Challenger - Benchmark)
No Difference (Benchmark = Challenger)
Cumulative Log-Likelihood Score (Higher is better)
Date (2024)
200
150
100
50
0
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01


Cumulative Predictive Likelihood: Stable_Volatility (t-1) -> Crypto_Returns (t)
Cumulative Log-Likelihood (Challenger - Benchmark)
No Difference (Benchmark = Challenger)
Cumulative Log-Likelihood Score (Higher is better)
Date (2024)
30
25
20
15
10
5
0
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01

GARCH (VaR):

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Final OOS VaR Backtest Results (5.0%)
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	Source_Factor	Target_Factor	OOS_Days	Expected_Breach_%	Benchmark_Breach_%	Benchmark_Kupiec_p	Copula_Breach_%	Copula_Kupiec_p
0	Stable_Volatility	Crypto_Returns	366	5.0	1.639344	6.425863e-04	6.830601	0.127009
1	Stable_Volatility	Crypto_Volatility	366	5.0	0.546448	7.440818e-07	4.644809	0.752464

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Final OOS VaR Backtest Results (1.0%)
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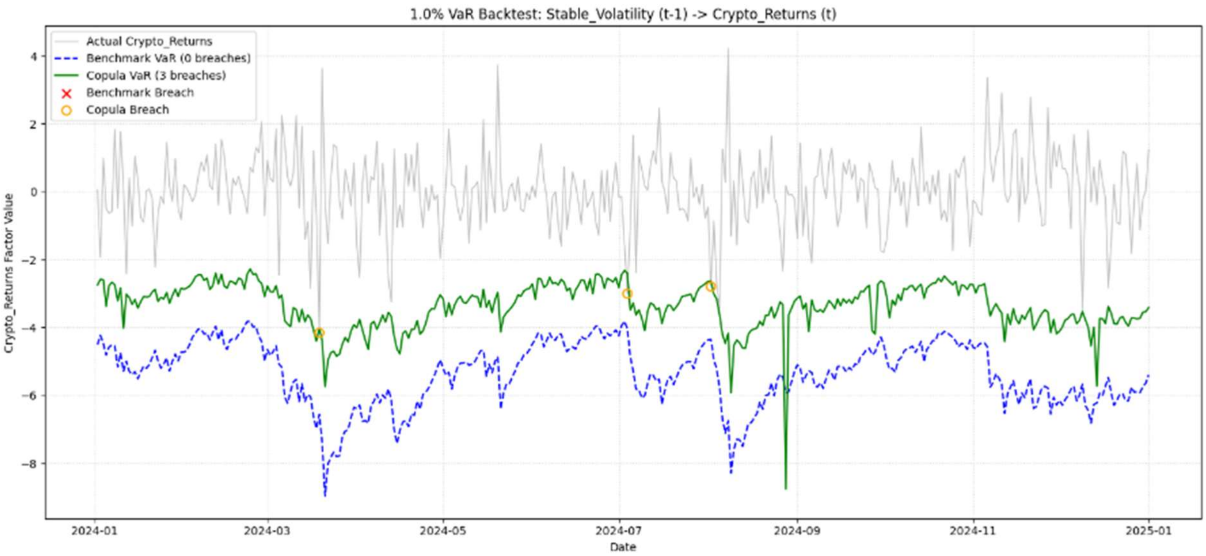
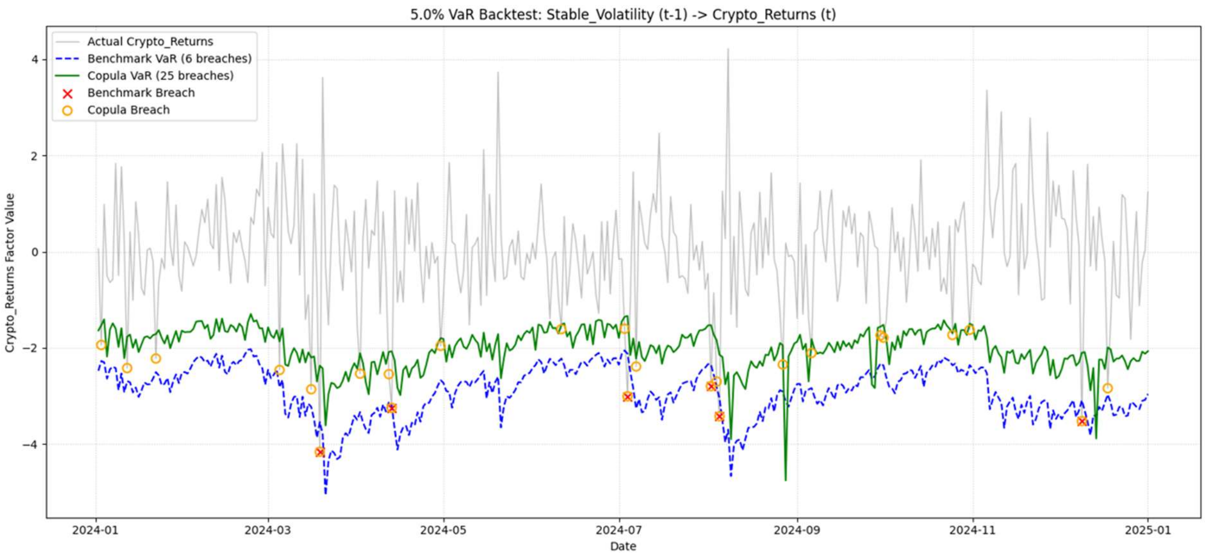
	Source_Factor	Target_Factor	OOS_Days	Expected_Breach_%	Benchmark_Breach_%	Benchmark_Kupiec_p	Copula_Breach_%	Copula_Kupiec_p
0	Stable_Volatility	Crypto_Returns	366	1.0	0.000000	NaN	0.819672	0.720414
1	Stable_Volatility	Crypto_Volatility	366	1.0	0.273224	0.097586	0.819672	0.720414





XGBoost:

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Final OOS Copula-ML (XGBoost) Backtest Results
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       Source_Factor      Target_Factor  OOS_Days  MSE_Benchmark  MSE_Challenger  Sharpe_Benchmark  Sharpe_Challenger
0      Stable_Volume     Crypto_Returns       366       0.054281        0.055014          0.281784           0.636751
1      Stable_Volume  Crypto_Volatility       366       0.014466        0.014309         -0.829453          -0.747475
2  Stable_Volatility     Crypto_Returns       366       0.054206        0.055396          0.824553           1.087865
3  Stable_Volatility  Crypto_Volatility       366       0.014131        0.013541         -0.755628           1.186501
```